Software Design Description (SDD)

Project: Pathfinder
Version: 2.0 of 2025
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6.1 Introduction

This document presents the architecture and detailed design for the Pathfinder project. Pathfinder is a career development web application that provides AI-driven resume analysis, job matching, career guidance, and learning resources to users navigating their educational and professional journeys.

6.1.1 System Objectives

Pathfinder aims to:

- Provide AI-powered career recommendations
- Analyze resumes and generate job-tailored cover letters
- Assist with job matching using external APIs
- Offer real-time AI chatbot support
- Present curated career resources from LinkedIn, Coursera, and others
- Engage users through personalized notifications and a dynamic dashboard interface

6.1.2 Hardware, Software, and Human Interfaces

Hardware Interfaces:

- User Devices: Desktop, laptop, tablet, and mobile
- Cloud Infrastructure: Google Cloud and Firebase for hosting, data processing, and load balancing

Software Interfaces:

- React.js: Responsive UI development
- Node.js: Backend processing and logic
- Firebase: Authentication, data storage, real-time updates
- OpenAI API: AI chatbot, resume feedback
- External REST APIs: Job listings and course data

Human Interfaces:

- Web GUI: Cross-platform accessible via modern browsers
- Chatbot: Floating interface active on all main pages
- Responsive Layout: Touch-friendly mobile and tablet views

6.2 Architectural Design

6.2.1 Major Software Components

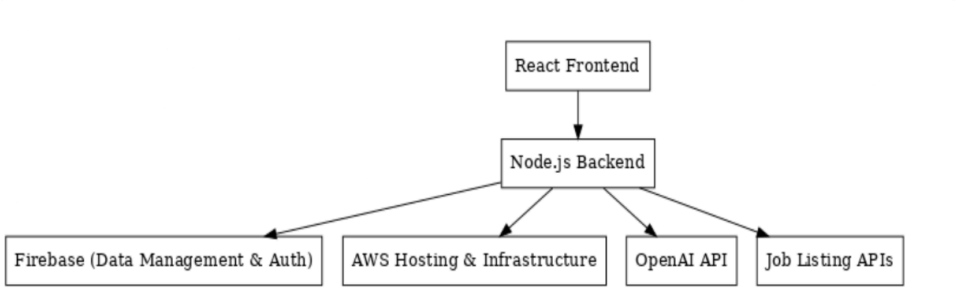
1. **Authentication Module**: Firebase login, session control, account recovery
2. **Dashboard Module**: Main user interface featuring widgets for each major feature
3. **Resume Analyzer**: AI-driven resume evaluation and feedback system
4. **Job Matcher**: Integrates job board APIs and filters results
5. **AI Chatbot Module**: Real-time OpenAI interactions, resource linking
6. **Notification Engine**: Sends alerts, tips, resume reminders

6.2.2 Major Software Interactions

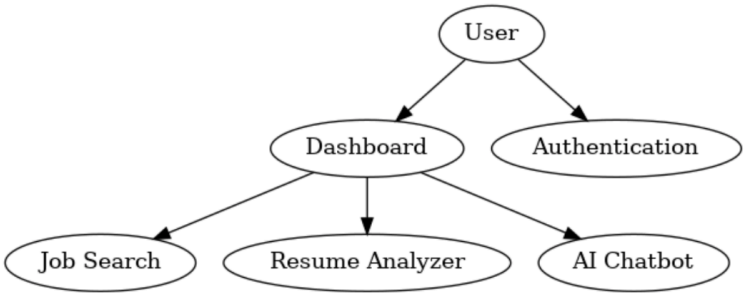
- **React** ↔ **Node**: REST APIs for passing user inputs and fetching results
- **Node** ↔ **Firebase**: Auth and Firestore SDK for secure data handling
- **Node** ↔ **OpenAI**: AI API calls with error handling and context management
- **Node** ↔ **External APIs**: Aggregates data from multiple job and education platforms

6.2.3 Architectural Design Diagrams

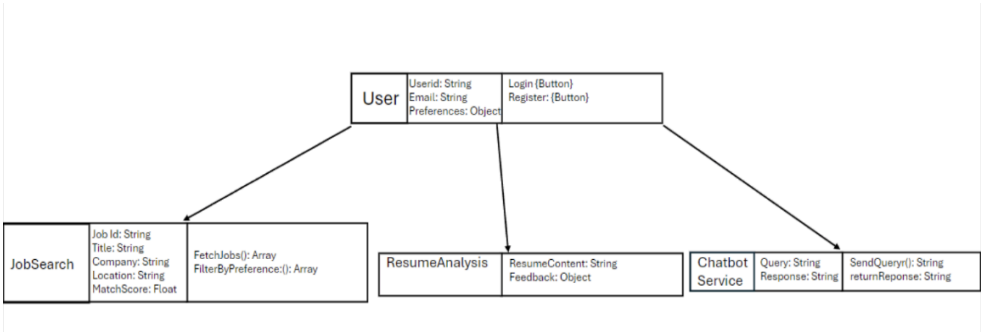
- Component Diagram



- Sequence Diagram



- Class Diagram



6.3 CSC and CSU Descriptions

6.3.1 Class Descriptions

6.3.1.1 User

- Fields: `userId`, `email`, `preferences`, `resumeId`
- Methods: `login()`, `register()`, `updateSettings()`

6.3.1.2 ResumeAnalyzer

- Fields: `resumeText`, `keywords`, `aiFeedback`
- Methods: `analyzeResume()`, `highlightWeakAreas()`

6.3.1.3 JobSearch

- Fields: `searchQuery`, `filters`, `results`
- Methods: `fetchListings()`, `applyFilters()`

6.3.1.4 ChatbotService

- Fields: `inputQuery`, `response`, `timestamp`

- Methods: `handleQuery()`, `fetchResponse()`

6.3.1.5 NotificationManager

- Fields: `type`, `targetUserId`, `message`
- Methods: `createNotification()`, `sendNotification()`

6.3.1.6 PresentationModule *(New)*

- Fields: `slideContent`, `imageAssets`
- Methods: `renderDemoSlides()`, `exportPoster()`

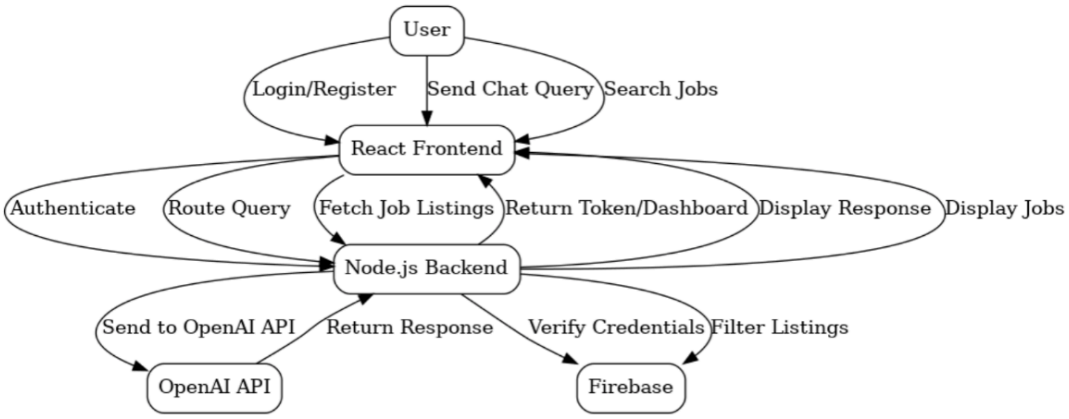
6.3.2 Interface Descriptions

- **Frontend** ↔ **Firebase**: Auth flow, real-time data sync
- **Backend** ↔ **OpenAI**: Chat message handling, prompt formatting
- **Backend** ↔ **External APIs**: RESTful calls for job/course data

6.3.3 Data Structure Descriptions

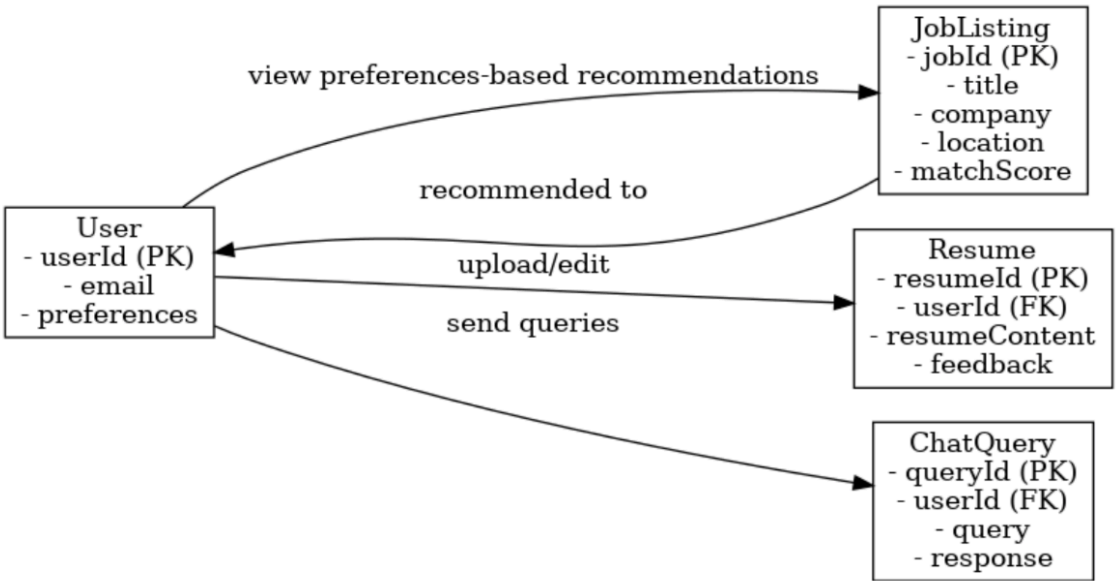
- `UserProfile`: `{ uid, email, preferences, linkedIn }`
- `Resume`: `{ id, content, score, suggestions[] }`
- `JobListing`: `{ id, title, company, matchScore }`
- `Notification`: `{ id, message, date, readStatus }`
- `ChatLog`: `{ sessionId, userQuery, aiResponse, timestamp }`

6.3.4 Design Diagrams



6.4 Database Design and Description

6.4.1 Database Design ER Diagram



6.4.2 Database Access

- CRUD via Firebase SDK (auth-restricted)
- Indexed access for quick query results

6.4.3 Database Security

- Access scoped via `uid` and roles
- HTTPS enforced globally
- Two-factor auth available on account settings